이론시험 문제
해설

주의. 각 채점 항목은 부분 점수를 주지 않습니다!

문제 1 (10.0점)

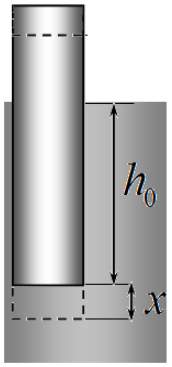
문제 A (3.0점)

A1. x

. 2

$$ma = mg - \rho S_0 (h_0 + x) g \,. \qquad (1)$$

m , ρ .



$$mg = \rho S_0 h_0 g \,. \qquad (2)$$

$$a = -\frac{g}{h_0} x \,. \qquad (3)$$

$$T = 2\pi \sqrt{\frac{h_0}{g}} \,. \qquad (4)$$

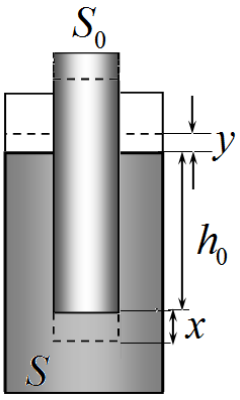
A2.1 x

$$\Delta U_1 = -mgx \,. \qquad (5)$$

x .

y

$$S_0 x = (S - S_0) y \quad \Rightarrow \quad y = \frac{S_0}{S - S_0} x \,. \qquad (6)$$



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$$\Delta m = \rho S_0 x \,, \qquad (7)$$

$$\Delta h_C = h_0 + \frac{1}{2}(x + y) = h_0 + \frac{1}{2}\left(x + \frac{S_0}{S - S_0} x \right) = h_0 + \frac{1}{2} \frac{S}{S - S_0} x \,. \qquad (8)$$

$$\Delta U_2 = \Delta m g \Delta h_C = \rho S_0 x g \left(h_0 + \frac{1}{2} \frac{S}{S - S_0} x \right). \qquad (9)$$

(2)

:

$$\Delta U = \Delta U_1 + \Delta U_2 = \frac{1}{2} \frac{S_0 S}{S - S_0} \rho g x^2 \,. \qquad (10)$$

A2.2 v_0

$$v_0 S_0 = v (S - S_0) \quad \Rightarrow \quad v = \frac{S_0}{S - S_0} v_0 \,. \qquad (11)$$

